



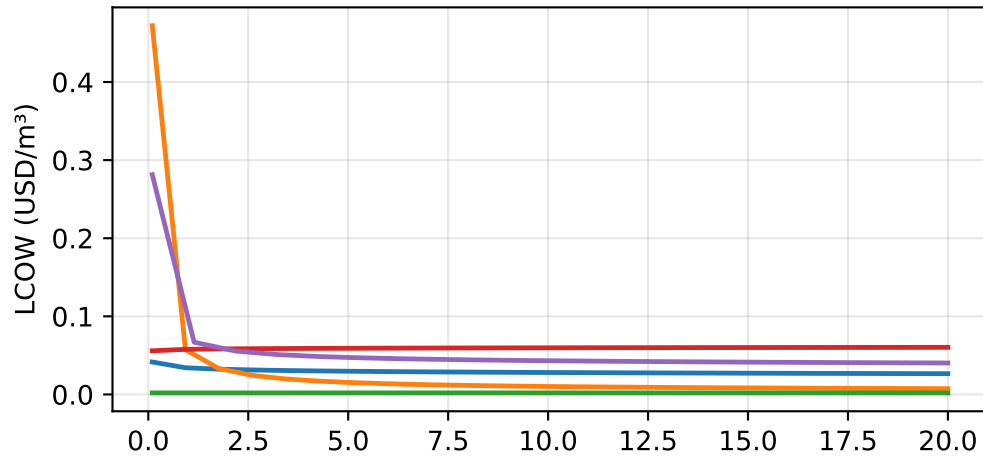
WaterTAP Zero-Order Model LCOW & CAPEX Curves

All costs in 2023 USD

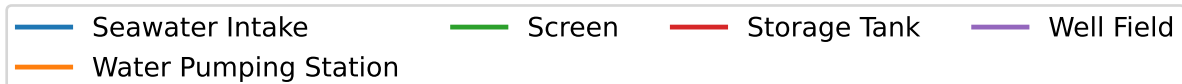
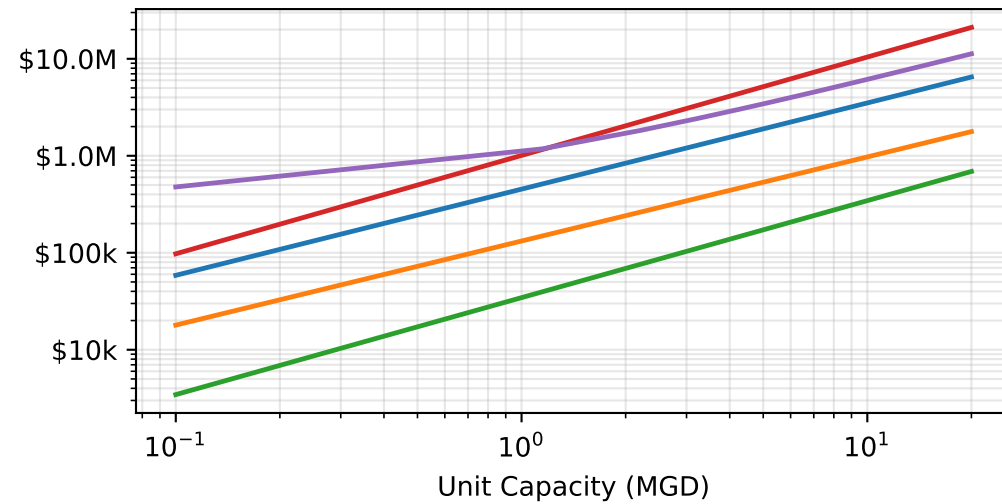
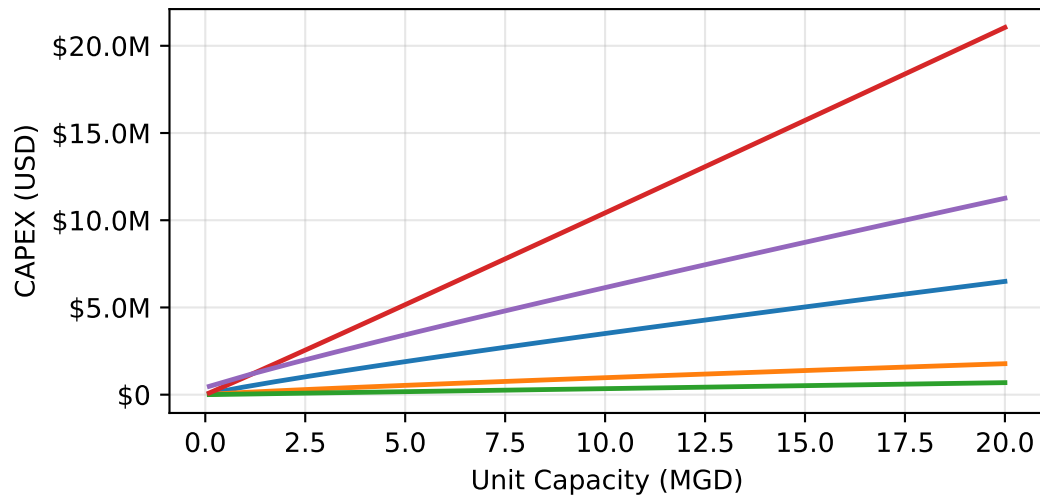
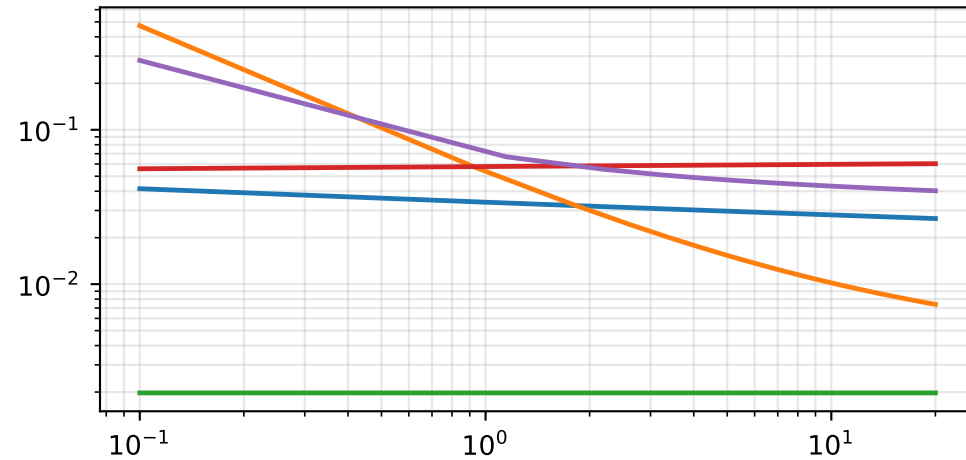
Generated from model runs

Intake & Conveyance

Linear Scale



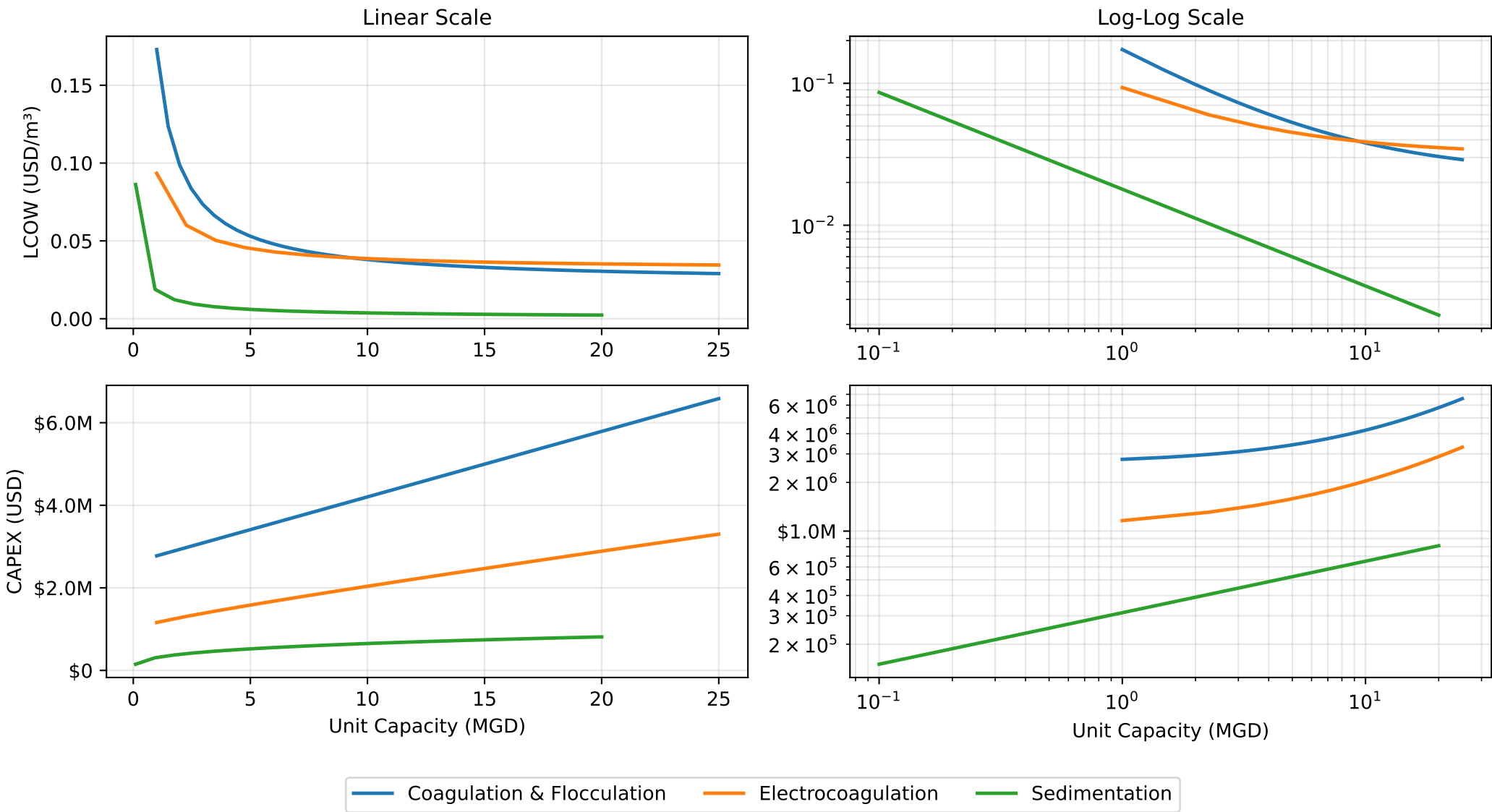
Log-Log Scale



Details:
Well field results assume 1 mile of pipe distance.



Pretreatment

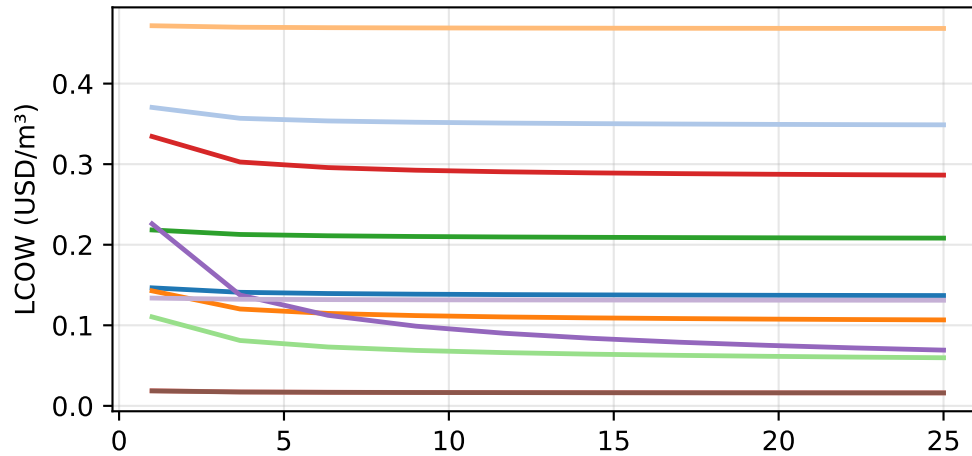


Details:

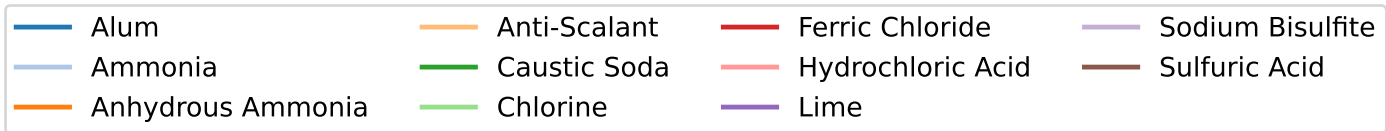
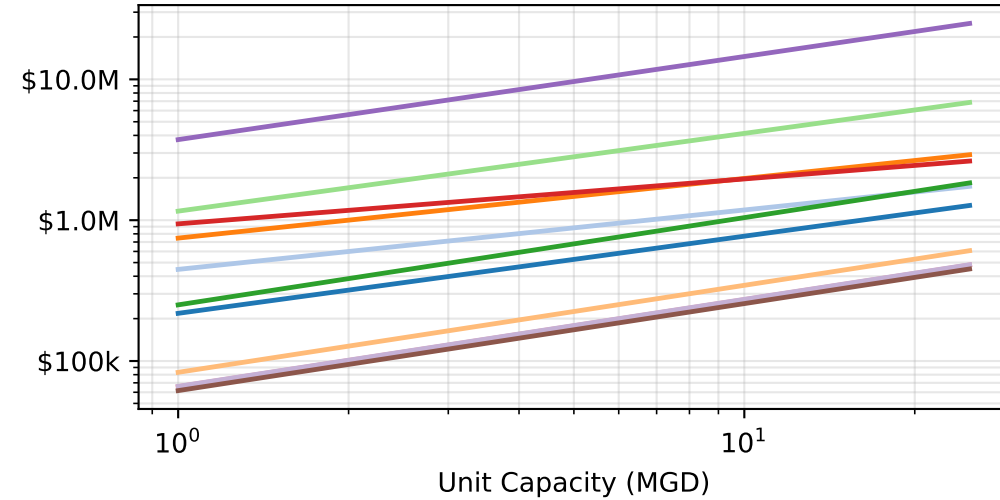
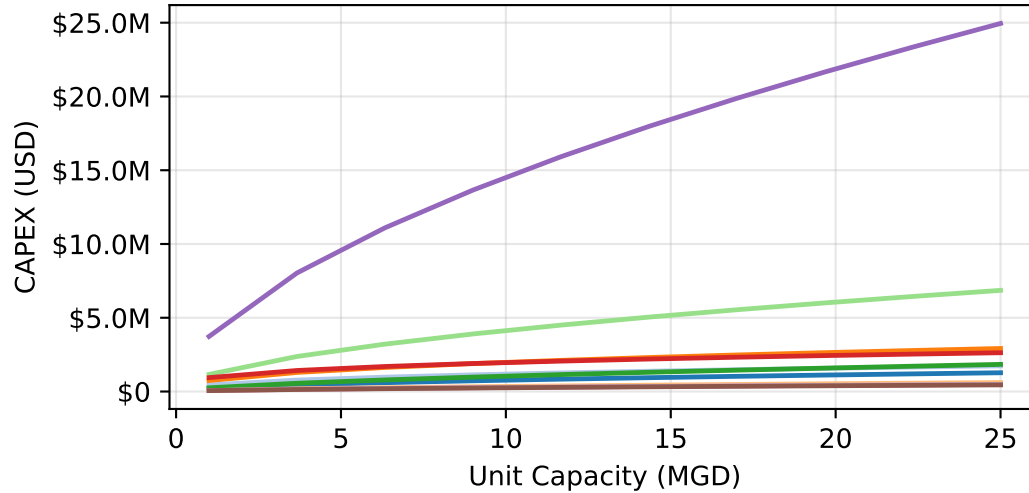
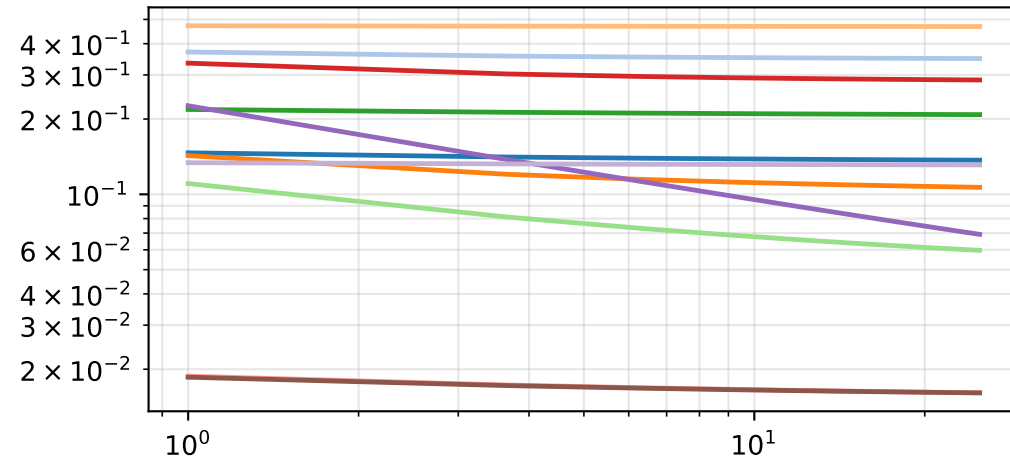
Electrocoagulation assumes a TDS of 1000 mg/L, a metal dose of 10 mg/L, and uses aluminum as the electrode material. Coagulation and flocculation are assumed to use 10 mg/L of alum and 0.1 mg/L of polymer.

Chemical Addition

Linear Scale



Log-Log Scale

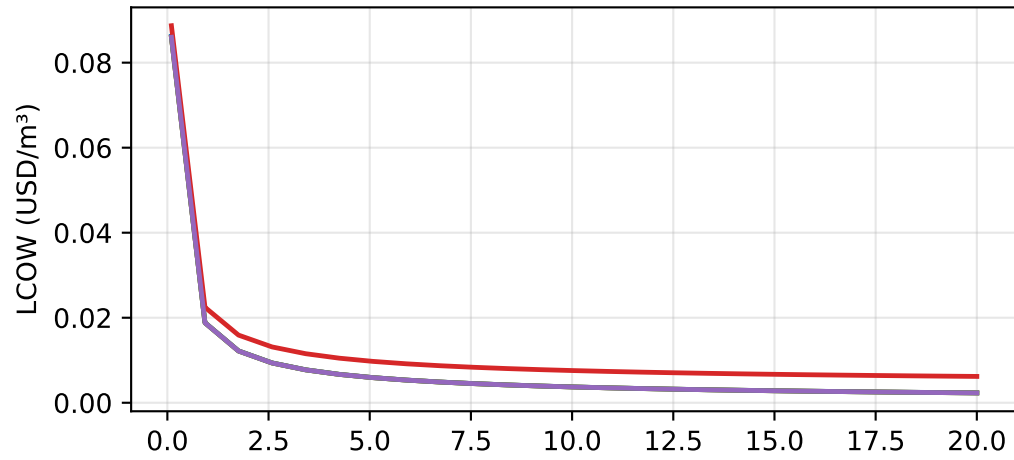


Details:
Chemical addition costs are run with a fixed dose of 100 mg/L for each chemical

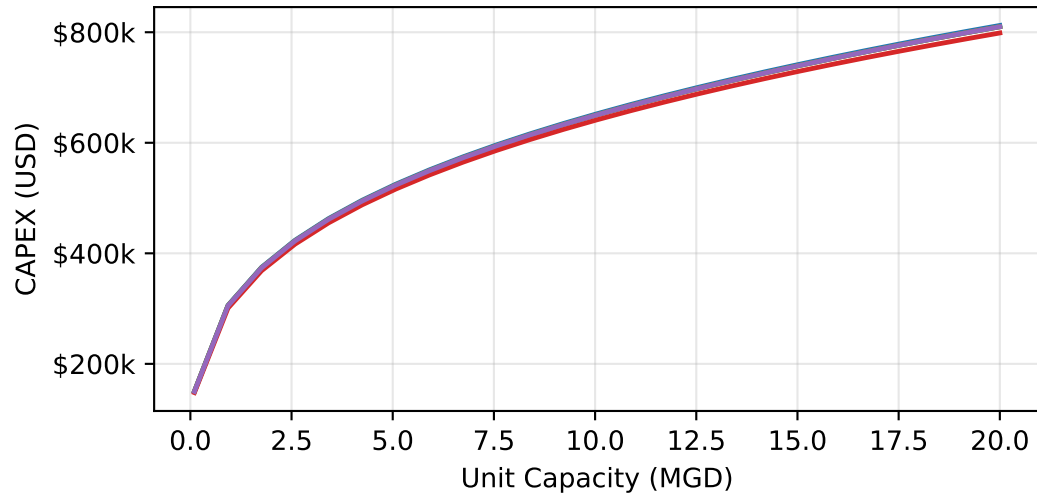
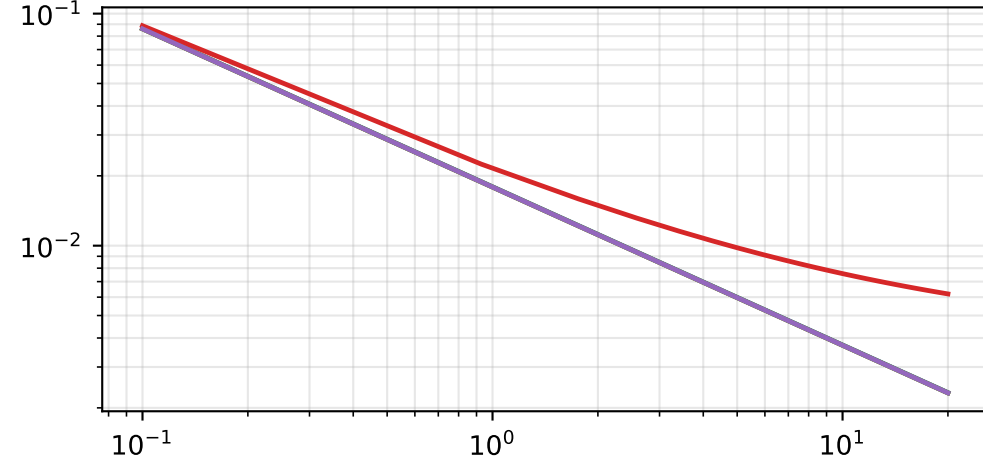


Clarification & Settling

Linear Scale

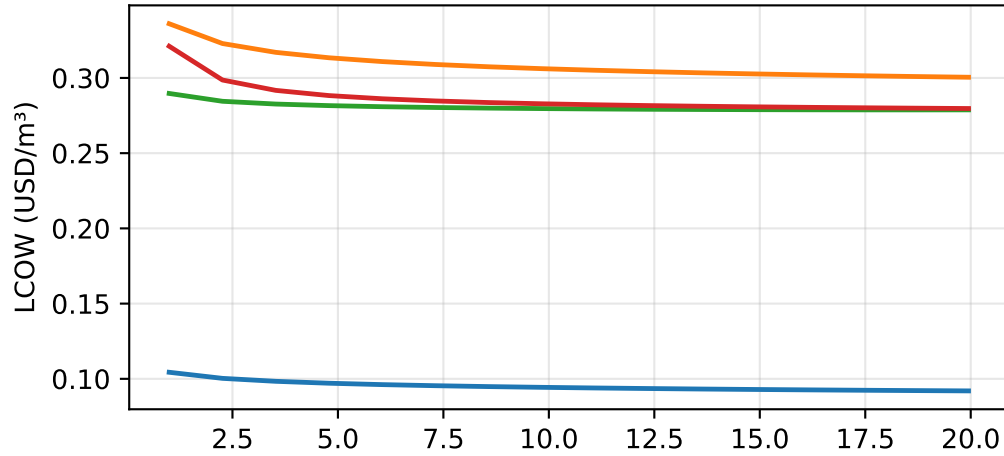


Log-Log Scale

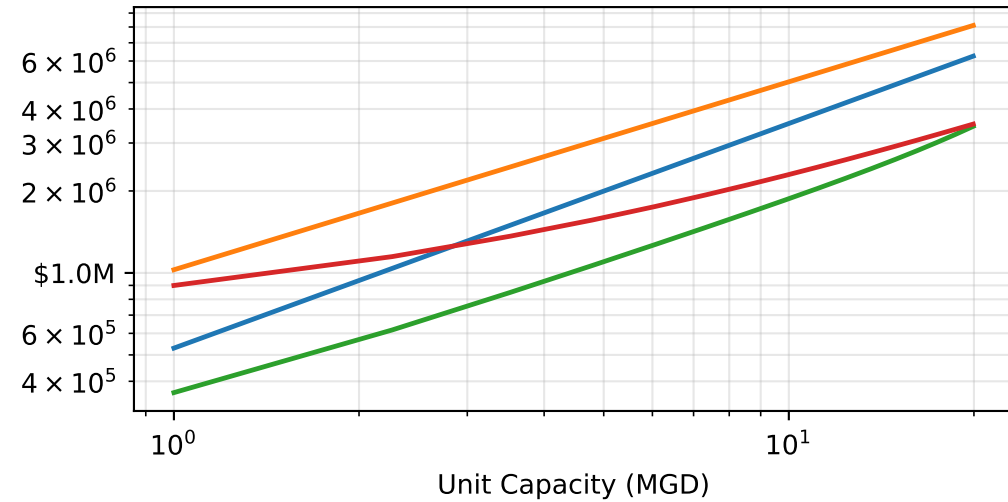
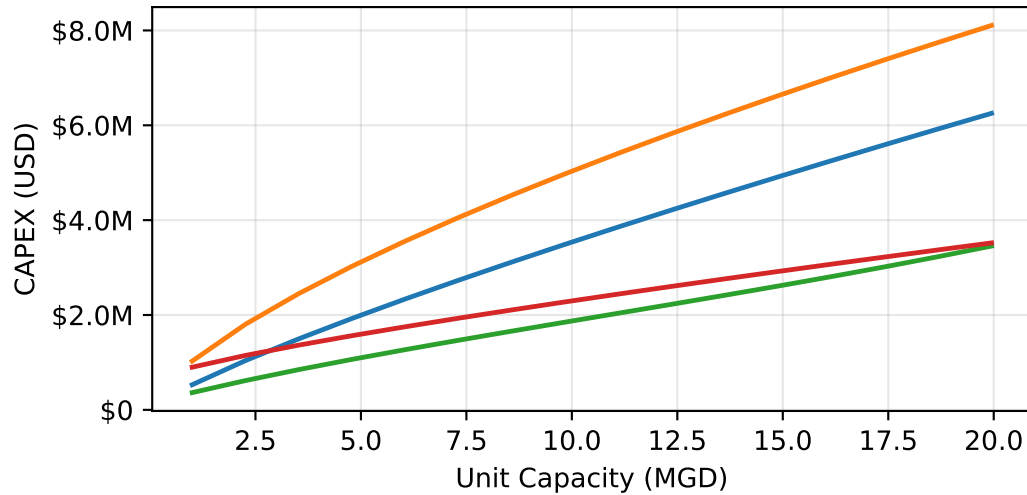
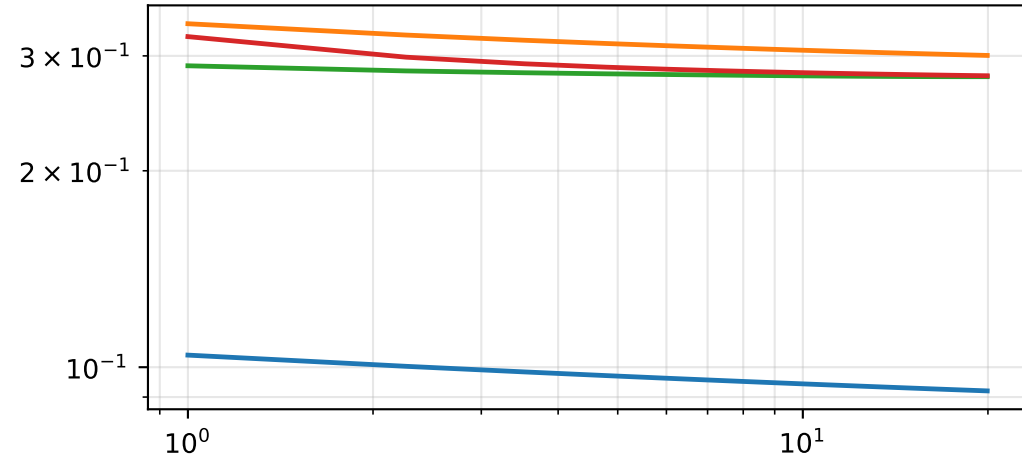


GAC & Ion Exchange

Linear Scale



Log-Log Scale



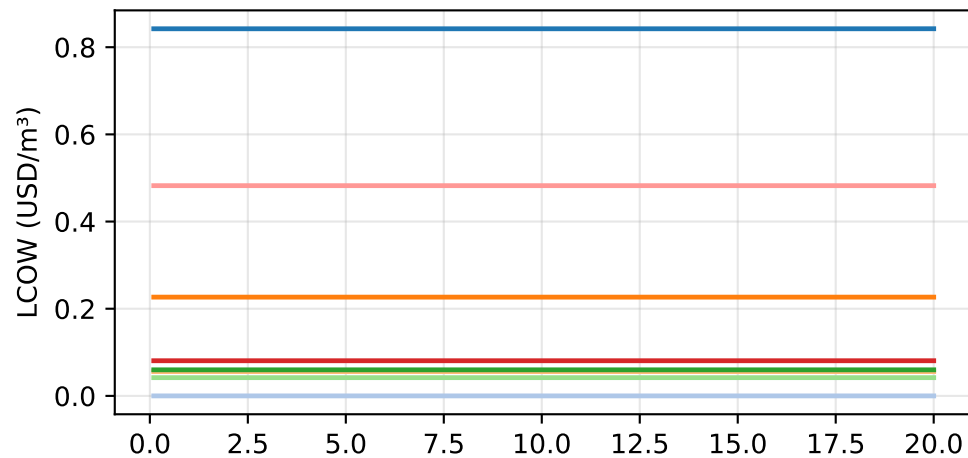
— Anion Exchange — Cation Exchange — GAC - Pressure Vessels — GAC - Gravity Basins

Details:

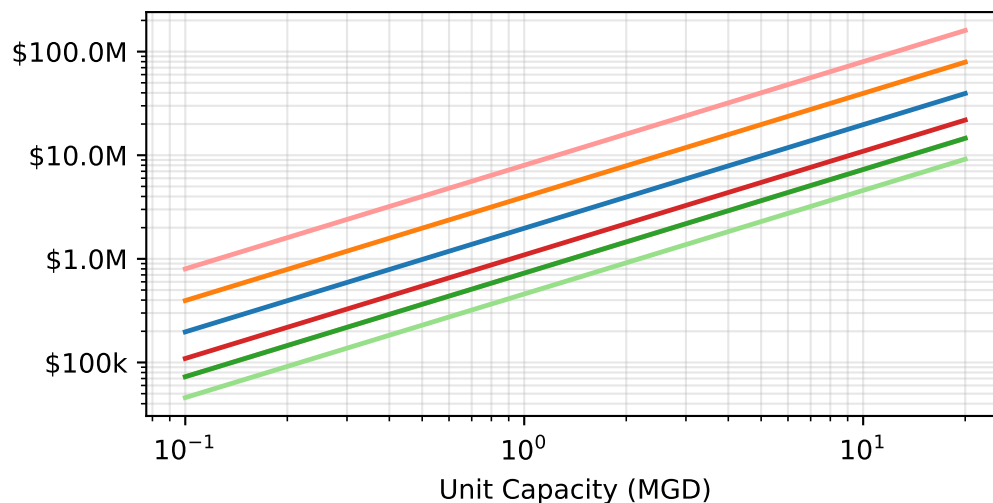
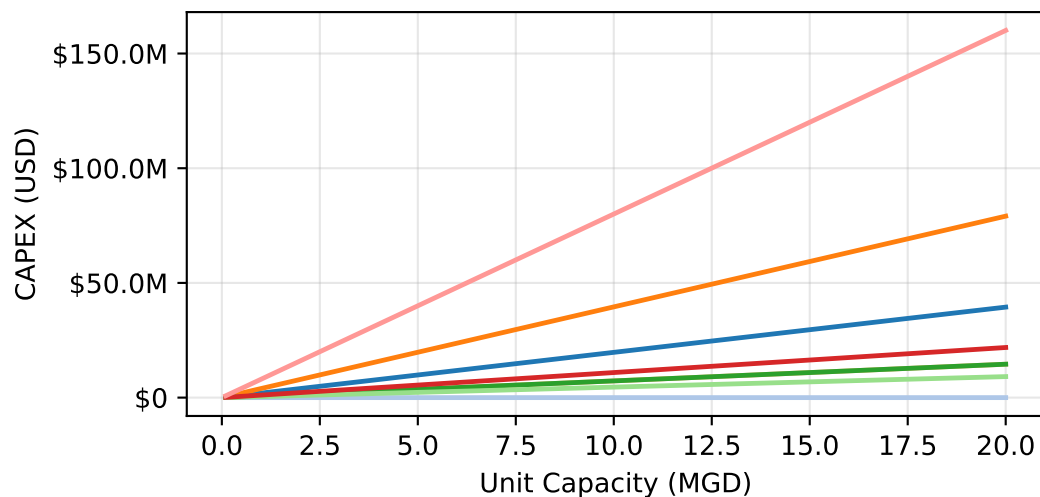
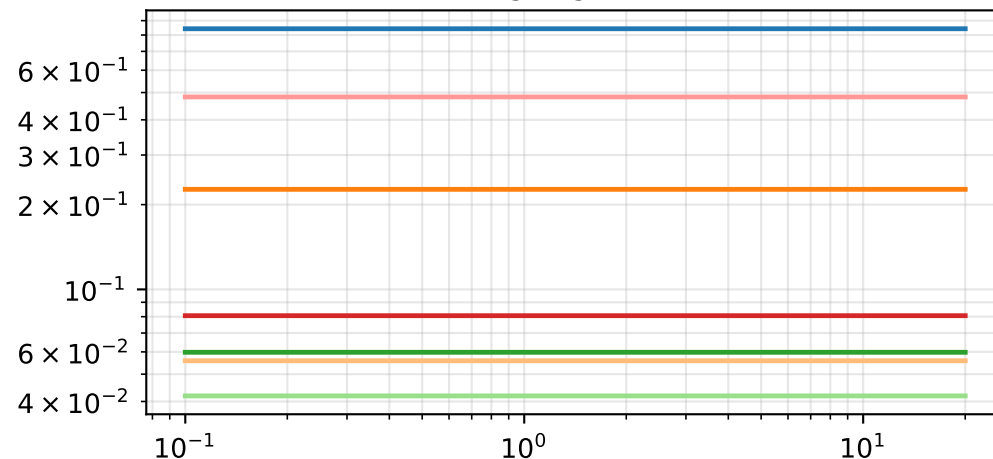
Anion exchange assumes 50 mg/L influent SO₄ concentration.
 Cation exchange assumes 200 mg/L inlet hardness concentration.
 GAC assumes 10 minute EBCT.

Membrane Processes

Linear Scale

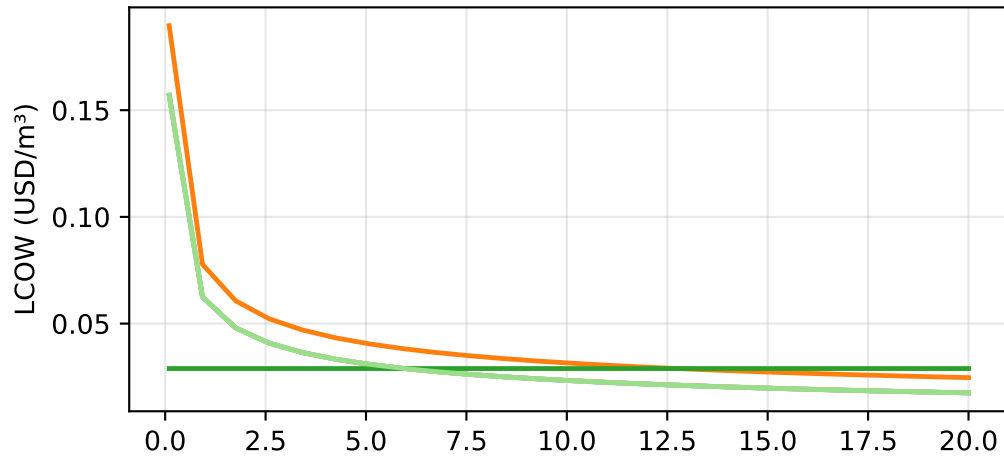


Log-Log Scale

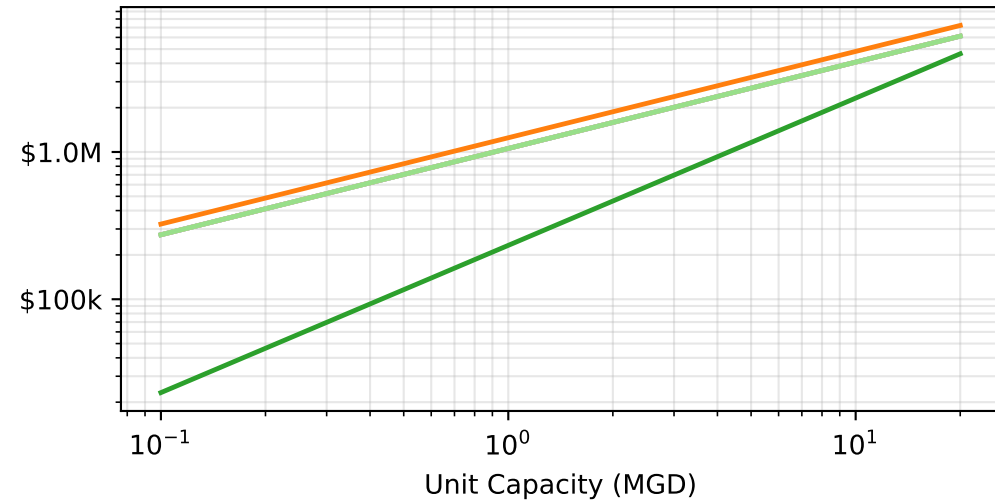
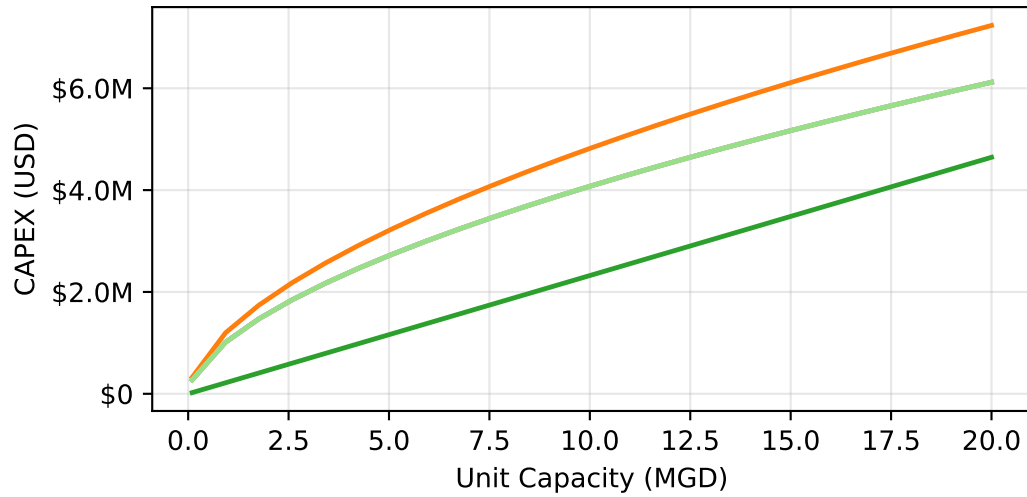
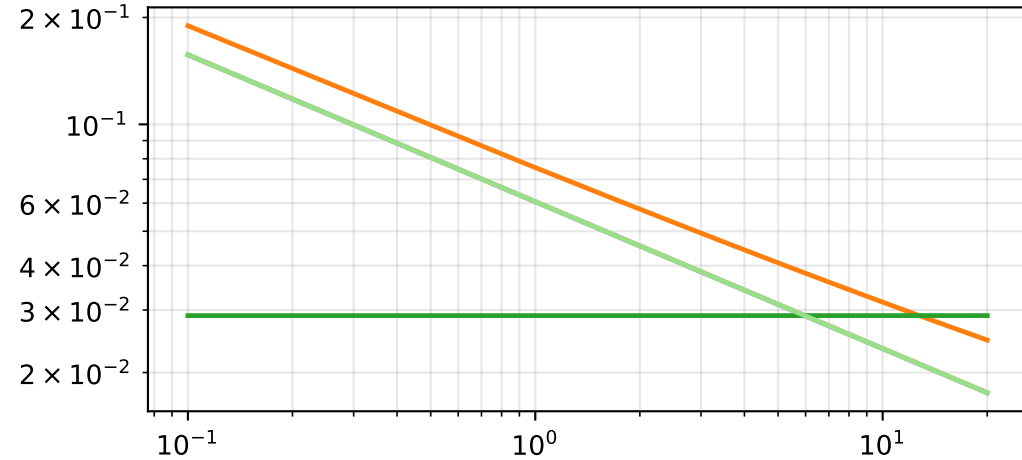


Filtration Technologies

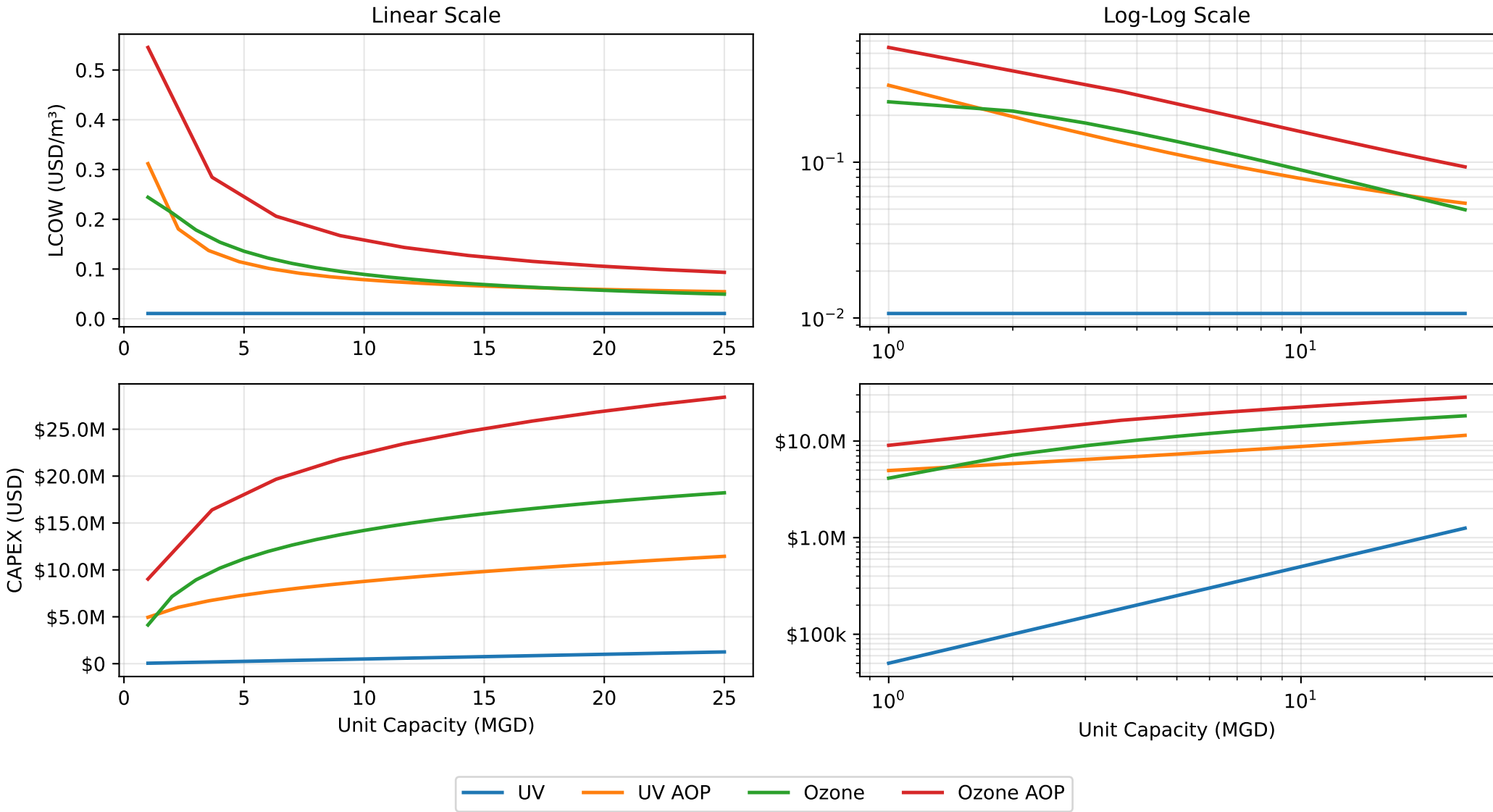
Linear Scale



Log-Log Scale



UV & Ozone AOP

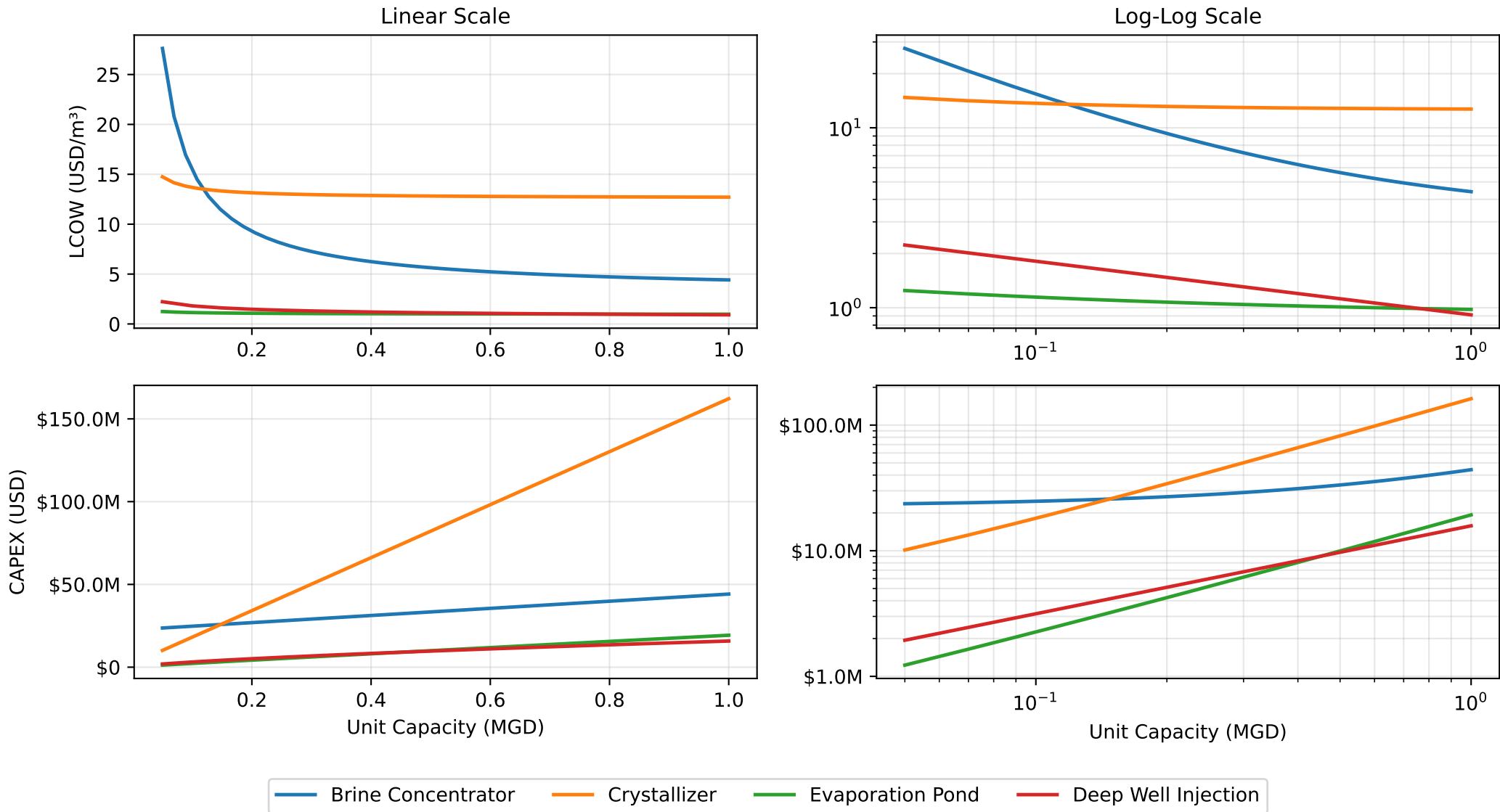


Details:

UV and ozone AOP costs are run with a fixed oxidant dose of 10 mg/L and UV SEC of 0.1 kWh/m³.
Ozone costs are for a fixed ozone dose of 9 mg/L.



Brine Management



Details:

Brine concentrator and crystallizer CAPEX and SEC are a function inlet flow rate and salinity.

These results are for a feed salinity of 60,000 mg/L TDS for the brine concentrator and 145,000 mg/L TDS for the crystallizer.

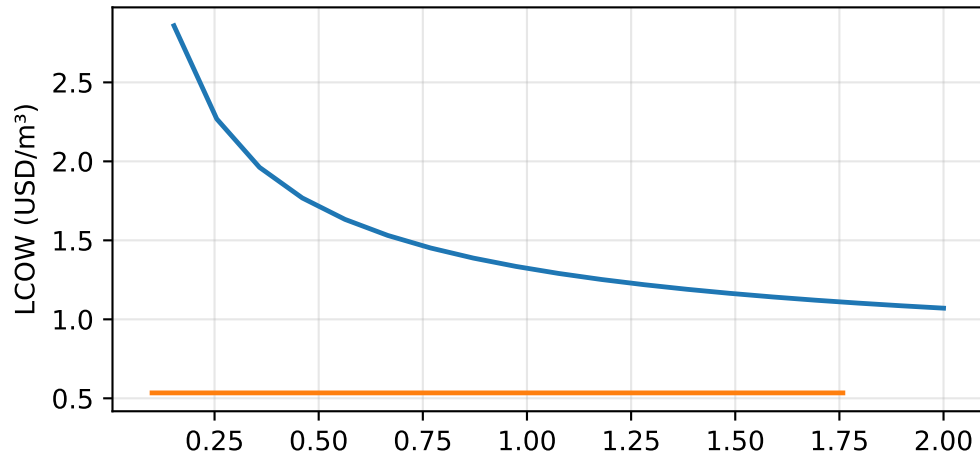
Evaporation pond costs are a function of inlet flow rate and solar radiation, with results shown for 25 MJ/m²/day.

Deep well injection costs are a function of inlet flow rate and pipe distance, with results shown for a pipe distance of 25 miles.

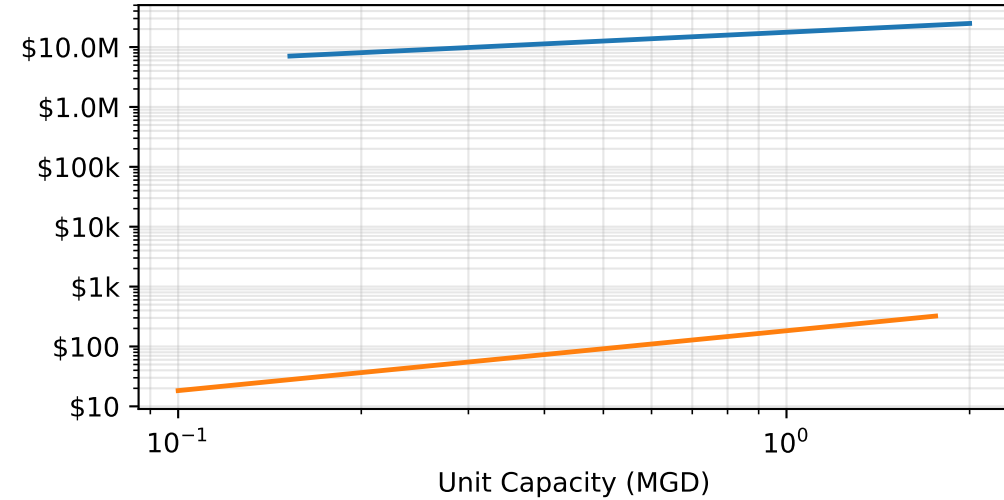
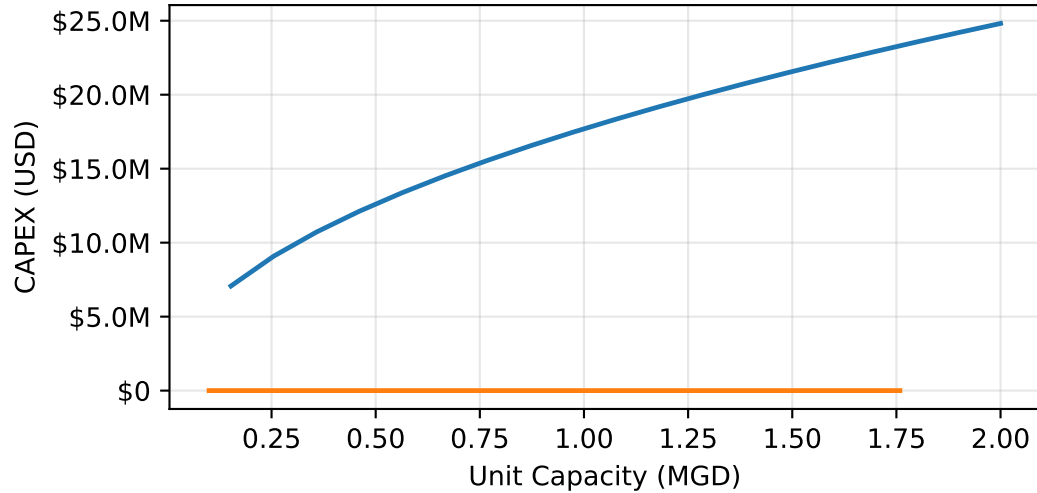
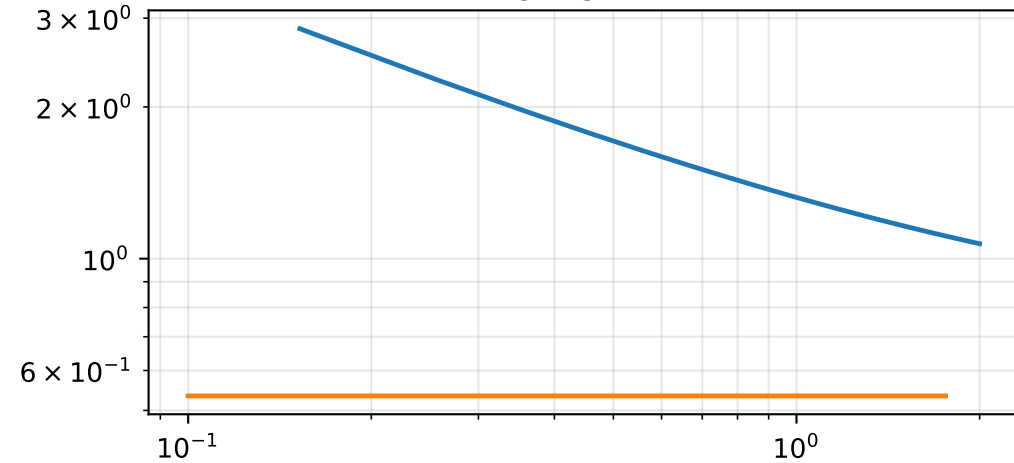


Solids Handling

Linear Scale



Log-Log Scale



Filter Press Centrifuge